MODULE 7.1

Virtualization and Cloud Computing

Hypervisors | Cloud Models | Cloud Security Threats

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The Virtual Environment

- **Virtualization:** separates the operating system from the underlying hardware using a hypervisor, letting multiple independent virtual machines (VMs) run on a single physical server.
- **Virtual network infrastructure:** each VM gets its own virtual NIC, connected through a virtual switch inside the host — effectively a small network living entirely inside one physical machine, before it ever reaches the real network.
- **Why it matters:** consolidating many VMs onto fewer physical servers cuts hardware, power, and cooling costs, speeds up provisioning a new "server" to minutes instead of days, and makes isolated testing and disaster recovery far easier.

	Type 1 (Bare Metal)	Type 2 (Hosted)
Installs on	Hardware directly	A host operating system
Performance	Higher — closer to the hardware	Lower — extra OS layer overhead
Typical use	Enterprise data centers, production servers	Desktop testing, development, demos
Examples	VMware ESXi, Microsoft Hyper-V, Citrix XenServer	VMware Workstation, Oracle VirtualBox

Cloud-Based Technology

- **Virtualization makes the cloud possible:** cloud computing is built on top of virtualization — physical resources are virtualized and then offered to many customers over a network, instead of each customer owning dedicated physical hardware.
- **Separates the application from the hardware:** just as virtualization separates the OS from the hardware, cloud computing separates the application and its data from any specific physical machine, so it can be reached from anywhere with network access.
- **On-demand, self-service:** a defining trait of the cloud is that resources can be provisioned and scaled automatically, without an administrator having to rack new physical equipment first.
- **The network is the delivery mechanism:** everything the cloud offers is delivered over the network — which means a reliable, well-secured network is just as critical to the cloud as the virtualized resources themselves.
- **Elasticity:** cloud resources can scale up during demand spikes and scale back down afterward, so customers pay roughly for what they use rather than for permanently over-provisioned hardware.

Cloud Computing: Deployment Models

A deployment model describes who owns the infrastructure and who it's shared with:

Model	Description
Public	Services offered over the open internet to anyone who pays; infrastructure is owned and managed entirely by the provider (e.g., AWS, Microsoft Azure, Google Cloud).
Private	Infrastructure dedicated to a single organization, hosted either on-premises or by a third party — more control and security, at a higher cost than public cloud.
Hybrid	A mix of public and private clouds, connected so data and applications can move between them as needs change — e.g., bursting extra workload into the public cloud during peak demand.
Community	Infrastructure shared by several organizations with common concerns, such as security or compliance requirements — for example, several government agencies sharing one cloud.

Cloud Computing: Service Models

A service model describes how much of the stack the provider manages versus the customer:

Model	Provider Manages	Customer Manages
IaaS (Infrastructure)	Physical hardware, virtualization, storage, networking	Operating system, runtime, applications, data
PaaS (Platform)	Hardware + operating system + runtime environment	Applications and data only
SaaS (Software)	Everything — hardware through the application itself	Just their own data and how they use the application

Rule of thumb: IaaS → PaaS → SaaS, the provider manages progressively more of the stack, and the customer manages progressively less.

Top Threats to Cloud Computing

Threat	Why It Matters
Data breaches	Sensitive data exposed through weak access controls or a provider-side flaw — affecting every tenant sharing that infrastructure.
Insufficient identity & access management	Weak credential and identity controls let an attacker impersonate a legitimate cloud user.
Insecure APIs	Cloud services are provisioned and managed through APIs — a poorly secured one is a direct path into the whole environment.
Account hijacking	Stolen credentials hand an attacker full control of a cloud account, including every service and dataset inside it.
Malicious insiders	A current or former employee with legitimate access abuses it for data theft or sabotage.
Shared technology vulnerabilities	Multiple tenants share the same underlying hardware and hypervisor — a flaw there can cross tenant boundaries.
Data loss	Accidental deletion, a provider-side disaster, or lost encryption keys can make data unrecoverable, with no attacker required.
Denial of service	An attack — or even a traffic spike — that exhausts shared resources can degrade service for every tenant on that infrastructure, not just the target.

Summary

- **7.1.2 —** virtualization separates the OS from the hardware via a hypervisor; Type 1 (bare metal) runs in production data centers, Type 2 (hosted) runs on top of a desktop OS for testing.
- **7.1.4 —** the cloud is virtualization delivered over a network — resources are provisioned on-demand and scale elastically, which makes the network itself a critical part of cloud security, not just the virtualized resources.
- **7.1.5 —** deployment models (Public, Private, Hybrid, Community) describe who owns and shares the infrastructure; service models (IaaS, PaaS, SaaS) describe how much of the stack the provider manages versus the customer.
- **7.1.7 —** most top cloud threats trace back to identity and access — stolen credentials, insecure APIs, and insider abuse all grant unauthorized access, while shared-technology flaws and denial of service exploit the fact that infrastructure is shared across tenants.

References

Cisco Networking Academy — Network Defense, Module 7.1: Virtualization and Cloud Computing

https://www.netacad.com

Cloud Security Alliance — Top Threats to Cloud Computing

https://cloudsecurityalliance.org/research/top-threats

NIST — The NIST Definition of Cloud Computing (SP 800-145)

https://csrc.nist.gov/publications/detail/sp/800-145/final